📘 Class Note: Operators in T-SQL

🔹 1. Introduction

Operators in T-SQL are **symbols or keywords** that perform operations on one or more expressions. They are used for:

- Calculations
- Comparisons
- Filtering data
- Combining results

🔹 2. Categories of Operators

1. **Arithmetic Operators**
    - `+` (Addition)
    - `-` (Subtraction)
    - `*` (Multiplication)
    - `/` (Division)
    - `%` (Modulo)
2. **Comparison Operators**
    - `=`, `!=`, `<>`, `<`, `>`, `<=`, `>=`
3. **Logical Operators**
    - `AND`, `OR`, `NOT`
4. **Bitwise Operators**
    - `&` (AND), `|` (OR), `^` (XOR), `~` (NOT)
5. **Assignment Operator**
    - `=` (in `SET` or `UPDATE`)
6. **String Operators**
    - `+` (concatenation)
    - `LIKE` (pattern matching)
7. **Set Operators**
    - `UNION`, `UNION ALL`, `INTERSECT`, `EXCEPT`
8. **Special Operators**
    - `IN`, `BETWEEN`, `IS NULL`, `EXISTS`

🔹 3. Examples

✅ Basic (Arithmetic + Comparison)

```sql
-- Arithmetic
SELECT 15 + 5 AS Addition, 15 - 5 AS Subtraction,
       15 * 5 AS Multiplication, 15 / 5 AS Division;

-- Comparison
SELECT * FROM Employees WHERE Salary > 50000;
```

✅ Intermediate (Logical + String)

```sql
-- Logical
SELECT * FROM Employees
WHERE Department = 'IT' AND Salary > 60000;

-- String Concatenation
SELECT FirstName + ' ' + LastName AS FullName
FROM Employees;
```

✅ Advanced (Set + Bitwise + Special)

```sql
-- Set Operator
SELECT Name FROM Students
UNION
SELECT Name FROM Teachers;

-- Bitwise
SELECT 5 & 3 AS BitwiseAND, 5 | 3 AS BitwiseOR, 5 ^ 3 AS BitwiseXOR;

-- Special Operator
SELECT * FROM Orders
WHERE OrderDate BETWEEN '2023-01-01' AND '2023-12-31';
```

🔹 4. Classwork with Solutions

🟢 Basic

Q1. Write a query to calculate the total price of 10 items, each costing 250.

```
SELECT 10 * 250 AS TotalPrice;
-- Result: 2500
```

Q2. Retrieve all employees who earn less than 40,000.

```
SELECT * FROM Employees
WHERE Salary < 40000;
```

🟡 Intermediate

Q1. Display employees whose department is “Finance” **or** salary is above 70,000.

```
SELECT * FROM Employees
WHERE Department = 'Finance' OR Salary > 70000;
```

Q2. Concatenate first name and last name into a single column called `EmployeeName`.

```
SELECT FirstName + ' ' + LastName AS EmployeeName
FROM Employees;
```

🔴 Advanced

Q1. Use `UNION` to combine two tables: `Customers` and `Suppliers`, showing only distinct names.

```
SELECT CustomerName AS Name FROM Customers
UNION
SELECT SupplierName AS Name FROM Suppliers;
```

Q2. Find all orders where `Quantity` is **between 50 and 100**.

```sql
SELECT * FROM Orders
WHERE Quantity BETWEEN 50 AND 100;
```

Q3. Perform a bitwise `AND` operation on `12 & 5`.

```sql
SELECT 12 & 5 AS BitwiseAND;
-- Binary: 12 (1100), 5 (0101) → 0100 = 4
-- Result: 4
```

🔷 5. Real-Life Use Cases

- **Payroll Systems** → Arithmetic operators for salary computation.
- **HR Filtering** → Comparison + Logical operators to shortlist candidates.
- **Merging Reports** → Set operators (`UNION` , `INTERSECT`) to unify branch data.
- **Fraud Detection** → Bitwise operators for encoded transaction flags.